CHAPTER THREE

LEARNING



What is Learning?

Meaning of Learning

- Learning is a relatively enduring or permanent, change of behavior as a result of:
  - Experience or
  - Practice.
- There are **three** important elements in it:
  - It involves ***change of behavior***
  - It's a ***change due to practice/ experience***
  - The change is ***relatively permanent*** – it must last as fairly long period of time

The foll. Changes in Behavior are not Considered as Learning

- Reflex action: sneezing or eye blinking
- Changes due to maturation and growth
- Temporary behavioral changes such as:



- Changes due to ***fatigue***,
- ***Disease*** and ***Drug*** states



- Permanent changes due to:
  - **Brain** damage



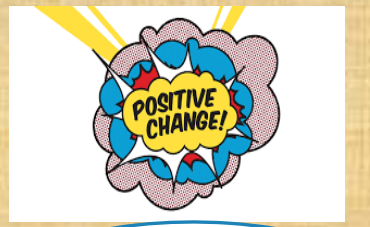
Characteristics of learning

- Learning is:
  - A continuous modification of behavior throughout life
  - Pervasive/universal, it reaches into all aspects of human life
  - Involves the whole person, socially, emotionally & intellectually
  - Often a change in the organization of experiences
  - Responsive to incentives
  - An active process
  - Purposeful; depends on maturation, motivation and multifaceted.



Cont…

- Learning is the product of environment: environment plays an important role in learning
- Learning affects the conduct of the learners:
  - *It could be* positively affecting OR
  - Negatively affecting peoples behavior



Negative change

➢ Principles of learning:

- Individuals learn best when they are:



- Physically,
- Mentally, and
- Emotionally ready to learn



Cont…

- Students learn best and retain information longer when they have:

  - Meaningful practice and exercise

- Learning is strengthened when accompanied by:

  - A pleasant or

  - Satisfying feeling, and

  - Is weakened:

  - When associated with an unpleasant feeling.



- Things learned best when:



  - A strong impression in the mind is created

  - It will be difficult to erase

Cont…

- Things most recently learned are best remembered
- The principle of intensity implies that

  - Students learn more from the real thing than from a substitute.

- Individuals must have some abilities and skills that may help them to learn
- Things freely learned are best learned

  - The greater the freedom enjoyed, the higher the intellectual and moral advancement

Factors Influencing Learning

- **Motivation:**

  - Motivation affects the effectiveness of learning.

  - The stronger and clearer the motives for learning, the greater are the effort to learn

  - When the motives of learning are high, learning becomes enthusiastic.

- **Intelligence:**

  - The more the individual is intelligent, the better she/he learns.



- **Maturation:**

  - Neuro-muscular coordination is important for learning a given task.

Example: The child has to be mature before she/he is able to learn.



- **Physical condition of the learner:**

  - The learner should be in:

    - A good health status to learn

Example: Sensory defects, malnutrition,

- Toxic conditions of the body, loss of sleep and

- Fatigue hinder effective learning.

- **Good working conditions :**



- **Absence of**:
  - Fresh air, light, comfortable surroundings,
  - Moderate temperature,
- **Presence** of distractions like noise and learning aids affect learning effectiveness.

- **Psychological wellbeing of the learner:**



- Individual's psychological states like:
  - Worries, fears, feelings of loneliness and
  - Inferiority hinders learning –
- Whereas self-respect, self-reliance, and self-confidence are necessary for effective learning.



Cont …

- **Background experiences:**



  - Factors such as:
    - Family background
    - Educational background

- **Length of the working period:**
  - Learning periods should neither be too short nor too long
  - I.e Long learning time sets:
    - Fatigue and
    - Reduces effectiveness in learning



Theories of learning

- **Behavioral Learning Theory**

  - Believes **learning occurs** as a result of **stimulus – response associations**

  - Emphasize on **observable behaviors** – explanations focus on their **consequences**

⇨ **There are two major types**:

- **Classical** conditioning

  - Focus on involuntary **emotional** or **physiological** responses such as:

    - **Fear, increased heartbeat**,
    - **Salivation or sweating**

    } Automatic Reponses to stimuli

Cont…

- Classical conditioning:

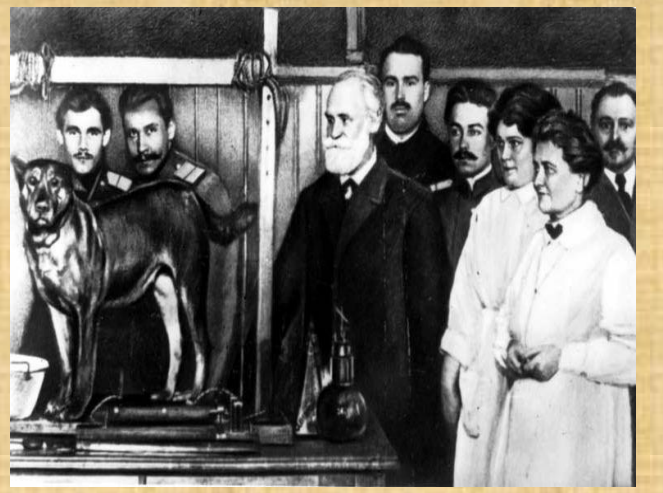
  - Is the procedure by which a **neutral stimulus** becomes a **conditioned** stimulus.

  - Ivan Pavlov was studying salivation in dogs as part of a research program on digestion.



- **Basics of classical conditioning**

Neutral Stimulus (NS)—Bell



- A stimulus that does **not normally** elicit a response by itself.

  I.e. before conditioning

  Example: bell ringing

Cont…

- **Unconditioned stimulus /UCS/:**

- A stimulus that reflexively and reliably elicit a response

Example: The meat



- **Conditioned stimulus (CS):**



- Any stimuli that elicit conditioned response as a result of paired with unconditioned stimulus.

Example: Bell

- **Unconditioned response (UCR):**

- Inborn response as a result of unconditioned stimulus.

Example: Saliva

- **Conditioned response /CR/:**



- which is elicited by the conditioned stimulus

Example: Salivation



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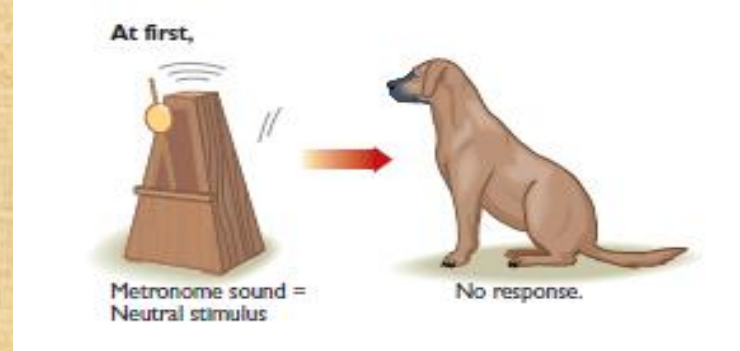
- Steps in classical conditioning:

Step 1: Before conditioning

Bell ringing ⟹ No response

(NS)

Meat (UCS) ⟹ Salivation (UCR)

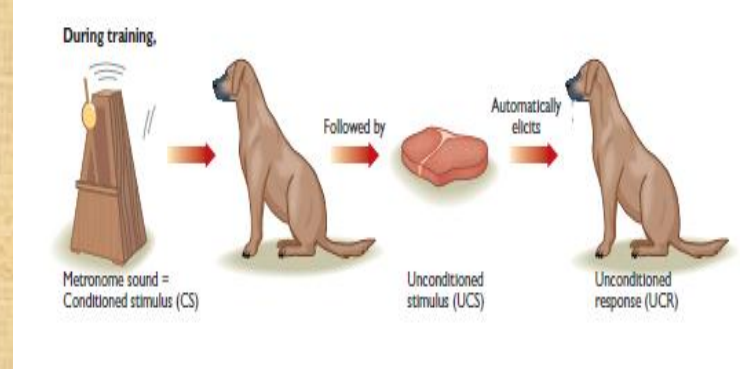


Step 2: During Conditioning

Bell + Meat ⟹ Salivation

(NS) + (UCS) UCR

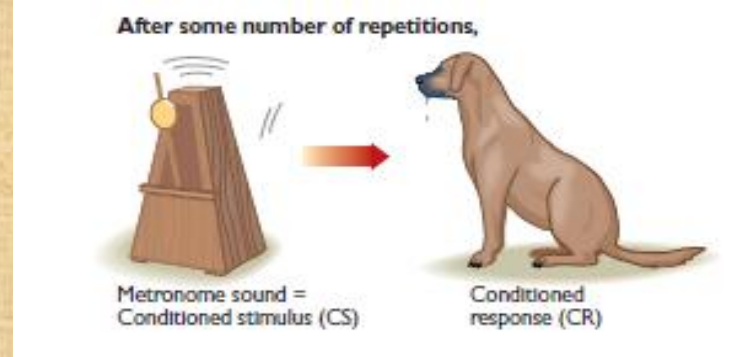
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Step 3: After Conditioning

Bell ⟹ Salivation

(CS) (CR)



Cont…

- **Principles of classical conditioning**:
  - ✓ **Acquisition**:

    - Previously neutral stimulus (bell) gets the ability to elicit response in the absence of US /Meat/.

  - ✓ **Stimulus generalization**:

    - When sbd, respond in similar way to all stimuli that looks like the conditioned stimulus.

    Example: phone ringing

Cont…

- Principles:
  - ✓ **Stimulus discrimination**:
    - Occurs when the individual responds differently to stimuli
  - ✓ **Extinction**:
    - The repeated presentation of the CS in the absence of UCS results in absence of CR.
  - ✓ **Spontaneous Recovery**:
    - After extinction a CR may suddenly reappear even without further conditioning trials.

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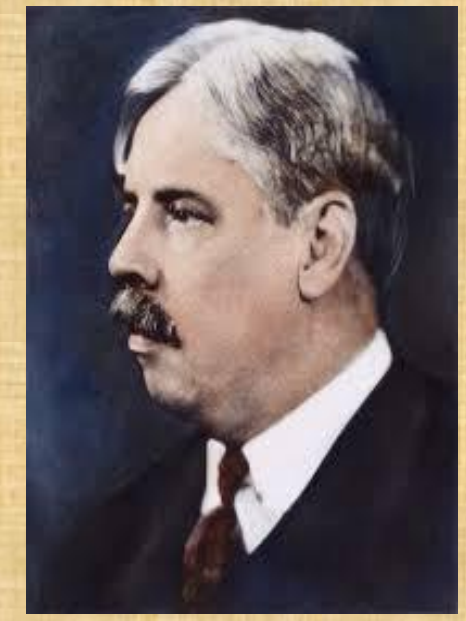
- Applications of classical conditioning:
  - Aversive conditioning: is a technique which involves unpleasant stimulation
    - Aims to induce such feelings as fear and dislike i.e. in relation to stimuli that trigger unwanted behavior.
  - Flooding: a technique for eliminating conditioned fears
    - The fear producing stimulus is presented by itself over and over again
    - i.e. confront the fear-eliciting stimulus – no escape

Cont…

- Systematic desensitization: this technique requires construction of a hierarchy of fears from least-feared to most feared.
- Advertising: A neutral product is associated with people, objects, or situations consumers to elicit a positive response
- Drug use: the CS associated with drug use lead individuals to seek out those substances

Cont…

- **Operant/ instrumental conditioning: /Thorndike/**
  - Learning occurs due to:



- Voluntary response depending on:

  - its **favourable** or **unfavourable** consequences

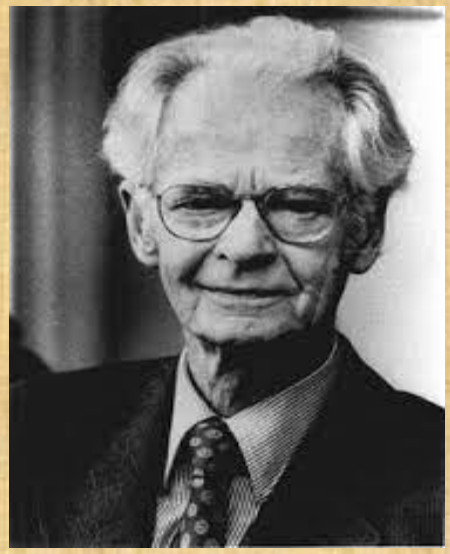
  I.e. Due to **reward** or **punishment**

- It increases or decreases the chance that the behaviour will **occur again** or **not**.

- ***Environmental consequences are given a focus***

Cont…

⇨ **B.F Skinner** further developed Thorndike's work in a more complex forms of behavior



- He argued that to understand behavior:
  - We should focus on the **external causes** of an action and the **actions consequences.**
    - I.e. we should look:
      - Out side the individual not inside

Cont…

- An emphasis on env'tal consequences is the heart of operant conditioning.

Classical conditioning	Operant conditioning
• Response is ***reflexive*** I.e ***natural*** and ***biological*** responses • An ***automatic*** response to sth. in the env't	• Response is ***complex*** and ***not reflexive*** • Response depends on ***environmental cues*** I.e. *punishment* & *reward* given

➢ There are two general forms of consequences that make to ***flourish*** or ***perish*** behavior:

Cont…

- There are **three types** of consequences:

  a. **A Neutral** Consequence – does not change the response

  b. **Reinforcement** – strengthens or makes it more likely to occur

- There are **two basic** types of reinforcers:

  - **Primary reinforcers:**

    - Food, water, air…

    i. e. those that satisfy our **biological needs**

    - Have the ability to ***reinforce with out prior learning***



Cont…

- … **types** of consequences:
  - **Secondary reinforcers**:

    - Reinforce due to their ***prior association with primary reinforcers***

    Eg. Money, good grade, praise, awards etc.

- Reinforcers may be:

  - Positive OR Negative



Cont…

- **Reinforcement**:

  - The process of increasing the probability of response to occur again in the future.

= Two types:

- **Positive reinforcement**: strengthens a response by ***presenting a satisfying*** event ***like getting freedom***.



E.g. if a good grade follows your hard work, your efforts to study are likely to continue or increase.

- Reinforcement …

- **Negative reinforcement**: involves the ***removal*** of ***sth. unpleasant***



E.g. If sbd. is nagging you to study, he stops when you study-

I.e you study to avoid nagging – usually called **escape learning.**

- **Avoidance Learning:** which refers to ***learning to avoid a painful, noxious stimulus prior to exposure***.

Example: Paying water and electric bills



Cont…

- **Schedules of reinforcement:**
  - **Fixed ratio schedules**: means that a reinforcement occurs only after **a fixed number** of responses are made by the subject.



Example: Employers are rewarded after a **fixed number** of productivity

  - **Variable ratio schedules**: means that a reinforcer is given after an **average number of correct responses**.

- Response is ***more resistant to extinction*** than ***fixed***

Example: Employers may be rewarded after producing 3, 4 or 5 cartoons of biscuits

Cont…

- **Fixed interval**: reinforcement occurs following ***the first response occurs after a fixed interval of time***



Example: When you chat, if you get a message after a 1min. time, then you check your message every minute.

- **Variable interval**: reinforcement of a response occurs only if a ***variable amount of time*** has passed ***since the previous reinforcer.***

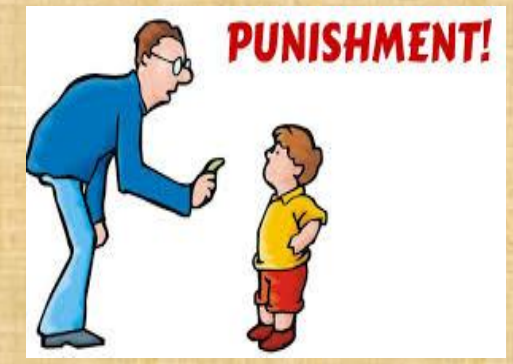
Example: When your message comes at 2,3,or 4 min. you check at variable time intervals

Cont…

c. Punishment: *weakens* or ***discourages*** the occurrence of certain behaviors or ***less likely to occur.***

=> Punishment has two types:

* ***Positive punish***.: refers to ***adding aversive (unpleasant)*** stimulus to get rid off the prevalence of unwanted behaviors.



Example: The teacher **warned** students when they disturb in the class



Cont…

* ***Negative punish***.: the process of ***removing pleasant events*** to avoid the probability of certain behavior that would likely to occur.



E.g: The teacher told students to leave the when they disturb.

➢ **Principles of punishment**

* It should immediately follow the behavior to be punished - **Immediacy**
* It should be consistent in its intensity, i.e. has to be up to the level of the mistake made - **Consistency**

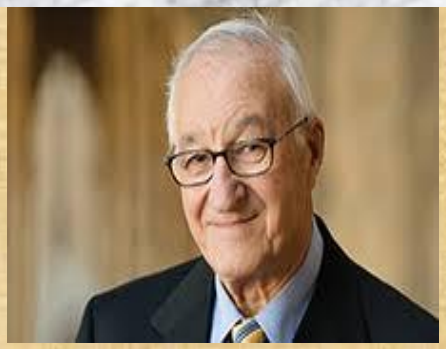
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- Being punished **intermittently/irregularly** becomes resistant to extinction
- Targeted only on the **unwanted behavior** but **not the person**
- Do not **mix punishment** with rewards for a given behavior
- Inform the individual **which behaviour you are punishing**

- **Punishment fails when:**
  - Inappropriately administered
  - The recipient responds with anxiety, fear and rage
  - Punishment is temporal/ impermanent
  - Most b/rs are hard to punish immediately
  - It conveys little message
  - It does not teach or show acceptable b/r
  - May foster hostility, aggression and passivity

* **Shaping:** is a procedure in which ***successively reinforcing*** behaviors that lead up to the ***approximate*** desired behavior.

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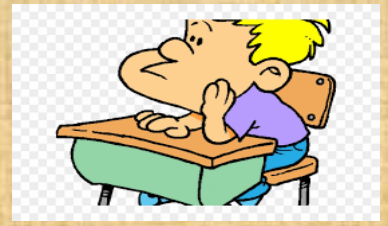
- **The social learning theory**:
  - Albert Bandura is considered as the father of this theory
  - He state that learning is acquired by:- **Imitation**
  - **Observing other people in social context** – the model
  - But not through standard conditioning procedures

- **Social cognitive theorists** believe that observational learning only **cannot** be fully understood without taking into account the **thought processes** of the learner
- They emphasize the **knowledge** that results when a person sees a model

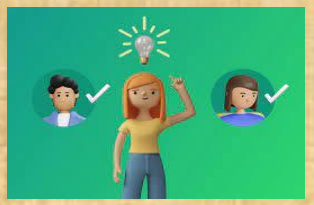
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= For Observational learning to occur:

- **Attention**: Paying attention to the models behavior and way of response



- **Retention**: recall or remember what has been happened.



- **Motor production**: Performing the observed behavior - i.e. putting on to practice



- **Motivation**: an interest to demonstrate what we have learned



Cont…

- **Cognitive learning theories**
  - **Latent/Hidden learning**:

    - learning that has occurred but is ***not immediately displayed*** or ***expressed*.**

    - learning that is ***not reinforced*** and ***not demonstrated*** until there is motivation to do so

    - It occurs without obvious **reinforcement** to be applied later.

Example: Hidden car driving skills

Cont…

- Insight/ Sudden learning:

  - States that learning occurs through a **sudden understanding of the relationship** of various parts of a problem.

  - *Providing sudden solution to a problem*

  - Rather than through trial and error